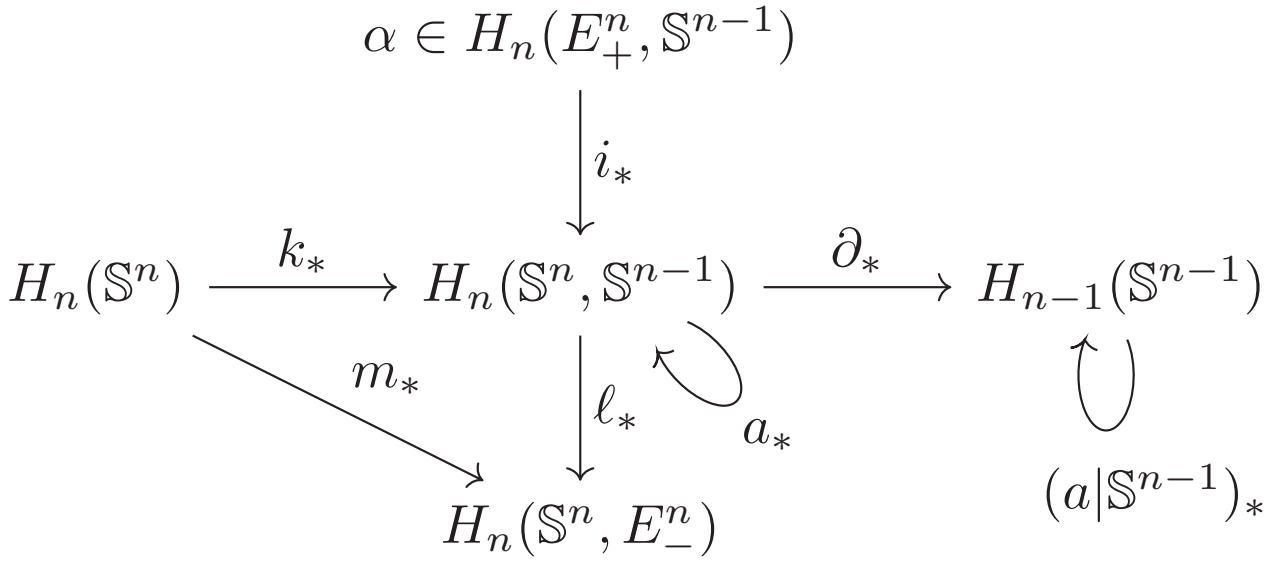

$\alpha \in H_n(E^n_+, \mathbb{S}^{n-1})$
i_*
$H_n(\mathbb{S}^n)$
k_*
$H_n(\mathbb{S}^n, \mathbb{S}^{n-1})$
∂_*
$H_{n-1}(\mathbb{S}^{n-1})$
m_*
ℓ_*
a_*
$(a|\mathbb{S}^{n-1})_*$
$H_n(\mathbb{S}^n, E^n_-)$